When Five AIs Agree with Me, Am I Closer to the Truth?

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Abstract Agreement from several AI systems is easily interpreted as independent confirmation. This paper examines when that interpretation is warranted. Combining a targeted synthesis of twelve primary studies, Bayesian evidence analysis, and reproducible probability constructions, it distinguishes output agreement, factual correctness, justified belief, and subjective confidence. The evidential value of consensus depends on its relative likelihood when a claim is true and when it is false. In a symmetric example with balanced priors and conditionally independent systems, each 80% accurate, unanimous agreement yields 99.90% accuracy. Introducing a common component while preserving individual accuracy can reduce this to 88.63%; complete duplication yields only 80%. These are analytical illustrations, not measurements of actual products. Pairwise correlation alone does not determine the reliability of unanimity. Sycophancy and selective reporting alter what agreement means, while discussion can transmit either useful information or a common error. The paper proposes retaining independent initial answers, tracing evidence to its sources, checking counterexamples, permitting abstention, and calibrating procedures on relevant tasks. It also outlines future studies that could challenge the framework. Agreement among five AIs can be evidence, but its incremental value comes from discriminating checks that contribute independent information, not from the number of endorsements alone.

Keywords: AI consensus; social epistemology; correlated errors; sycophancy; evidential independence; multiagent debate.

1. A reassuring but incomplete inference

Imagine proposing a claim and asking five AIs to assess it separately. The first endorses the argument. The second supplies supporting reasons. The third adds background. The fourth identifies minor qualifications but still agrees. The fifth concludes that your judgment is essentially correct. What reassures you is no longer just an individual reason: it is that they all say so.

This is a thought experiment, not a conversation collected for this paper. Yet the question it raises is precise: what is missing between “five systems agree with me” and “I am closer to the truth”?

We may treat five opinions as five checks, or as testimony from five experts. But five witnesses citing the same inaccurate report have not supplied five independent observations. Five calculators given the same mistaken formula establish consistency of calculation, not validity of the formula. AI systems present a more complicated case: they may contribute valuable independent reasoning, repeat shared knowledge, accept a common false premise, or adapt to the same user’s expectations.

The central argument of this paper is that **the reliability of consensus must be explained by how it is produced**. The relevant question is not simply how uniform the endorsements are, but whether equally uniform endorsements would occur if the claim were false. That counterfactual moves the discussion from reassurance to evidence.

The number five is not a special epistemic threshold. Three, five, and nine systems all face the same issue. More systems may reduce some incidental errors, but their number alone cannot eliminate shared blind spots. This paper assumes neither that majorities are necessarily correct nor that they are necessarily mistaken. It examines the conditions linking agreement to warranted trust.

2. Four different senses of getting closer to the truth

Four things must be distinguished. **Output agreement** means that systems reach the same conclusion. **Factual correctness** means that the conclusion satisfies a specified factual or formal criterion. **Justified belief** means that the reasons for accepting it withstand examination. **Subjective confidence** means that the user feels more certain. These may increase together, but they can also diverge.

Five systems may jointly guess the right answer without employing a reliable checking procedure. Conversely, a procedure may reasonably support a conclusion under limited information that later evidence overturns; this does not automatically make the procedure epistemically worthless. Assessing progress toward truth requires considering both whether the present answer is right and whether the procedure helps us make fewer mistakes across similar cases.

The main scope here is claims with assessable truth conditions and reasoning that can be checked. In prediction, today's consensus must await later outcomes and be assessed through calibration across cases. Questions such as whether a poem is moving or how one should live often combine factual claims, priorities, and personal circumstances. Similar AI advice may reflect similar value assumptions without revealing a uniquely correct value judgment. The binary probability models below do not capture the full significance of such questions.

"They all agree with me" also contains an easily overlooked variable: **my prior position has entered the question**. The user receives the evidence but may also shape its production. Asking neutrally whether a proposition holds supplies different information from saying that one believes it and requesting an assessment of one's argument. These conditions must enter the analysis if we want to know whether agreement responds to facts, reasoning, or expectations.

Finally, correctness requires a defensible criterion. Mathematical reasoning can be checked against premises and inference steps; programs can be tested; factual claims can be traced to original records. In annotation tasks with legitimate disagreement, even a human majority label is an operational standard. We cannot simply name human consensus "truth" and then use AI agreement with it as proof of truthfulness without discussing the standard's limits.

3. Method and prior research: retaining evidence on both sides

The paper combines targeted literature synthesis with theoretical constructions. Searches through October 26, 2026 covered correlated model errors, sycophancy, self-consistency, multiagent debate, explanation faithfulness, and human overreliance. Original papers, conference publication pages, and author preprints were preferred. The twelve studies were selected for their relevance to the argument; this is not an exhaustive systematic review, and effect sizes are not pooled across studies. The evidence ledger records the depth of each check. Some checks cover relevant full-text sections, while others cover only the original abstract and metadata. None of the reported model experiments was independently replicated here.

Evidence supporting collective reasoning must not be omitted. Wang and colleagues improved performance on the reasoning benchmarks they tested by sampling several reasoning paths and aggregating answers. Du and colleagues also reported reasoning and factuality benefits from multiagent debate on their tasks. These results reject the extreme claim that additional AI opinions can never help. They do not show that agreement among any five systems delivers the same benefit. [1], [2]

Other research clarifies the conditions of those benefits. Kim and colleagues observed correlated errors across a large collection of models; different providers or architectures did not guarantee independent errors. Choi and colleagues separated voting from discussion across seven NLP benchmarks and found that majority voting accounted for much of the improvement in their study. Claims about the effectiveness of debate therefore need a voting baseline with a comparable budget. [3], [4]

Sycophancy provides a third perspective. Sharma and colleagues found that users' expressed positions could influence the evaluations and answers of the assistants they tested, with human preference signals a possible contributing cause. This does not establish that all training with human feedback reduces truthfulness. TruthfulQA, meanwhile, made imitation of common human falsehoods an explicit evaluation target, illustrating that familiar statements and true statements can diverge. [5], [6]

Kohli's more recent preprint connects correlated errors in judge panels to effective sample size. It is treated here only as supplementary evidence. It addresses particular classification and preference tasks, and external peer-review status was not established in this review. Its panel measurements must not be treated as universal AI constants. [7]

The analyses actually completed for this paper are a derivation of the likelihood ratio of consensus; a five-system model that varies dependence while holding individual competence fixed; exact fractional calculations checked by enumerating all 32 binary voting patterns; and a counterexample with equal means and pairwise correlations but different reliability under unanimity. All subsequent numerical tables and figures are **analytical illustrations under stated assumptions**, not product evaluations.

4. When are five endorsements five pieces of evidence?

4.1 Compare agreement when a claim is true with agreement when it is false

Let H be the claim, A_i the event that AI i endorses it, and D the existing material, user position, prompting conditions, and known selection procedure. Bayes' rule gives:

$$\frac{P(H \mid A_1, ..., A_5, D)}{P(\neg H \mid A_1, ..., A_5, D)} = \frac{P(H \mid D)}{P(\neg H \mid D)} \frac{P(A_1, ..., A_5 \mid H, D)}{P(A_1, ..., A_5 \mid \neg H, D)}.$$

The final term is the likelihood ratio of consensus. Agreement increases the probability of the claim only if this ratio exceeds 1. If systems readily endorse both true and false propositions, endorsement may be pleasant but uninformative. If a mechanism produces agreement more often when the claim is false, the ratio can even fall below 1. That possibility needs specific evidence; distrust of AI alone does not establish it.

Define $a_i = P(A_i \mid H, D)$ as the probability that a true claim receives endorsement and $b_i = P(A_i \mid \neg H, D)$ as the probability that a false claim does. The joint likelihood ratio equals $\prod_i a_i/b_i$ **only when the answers are mutually independent conditional on D under both H and $\neg H$**. This conditional independence is the statistical requirement. Different product names are merely a possible clue.

Consider a deliberately simplified example: the prior is 50%; all five systems have $a_i = 0.8$ and $b_i = 0.2$; and their responses are conditionally independent. After five endorsements:

$$P(H \mid A_1, ..., A_5, D) = \frac{0.8^5}{0.8^5 + 0.2^5} = 99.90\%.$$

This result has a precise interpretation and strict limits. It is not an empirical rule that five AI endorsements warrant 99.90% confidence. Even with the same likelihood ratio, a prior of only 0.1% yields a posterior of 50.62%. The prior must reflect the current problem and available evidence; it cannot be selected to produce a preferred answer.

A single figure such as “80% accuracy” usually does not supply the necessary parameters. Performance may differ between true and false claims. If $a = 0.8$ but $b = 0.6$, the same balanced prior and conditional independence assumptions yield a posterior of 80.82% after five endorsements. The more readily a system endorses false claims, the less an endorsement distinguishes truth from falsehood.

4.2 Dependence permits updating, but not repeated counting

Independence is not necessary for consensus to be useful; it is necessary for the simple multiplication above. More generally, the joint likelihood ratio can be expanded by the chain rule. The incremental factor supplied by vote i is:

$$\frac{P(A_i \mid H, D, A_1, ..., A_{i-1})}{P(A_i \mid \neg H, D, A_1, ..., A_{i-1})}.$$

If the last four systems simply copy the first answer, subsequent endorsements are certain under either truth state once the first vote is known. Their incremental factors are all 1. In the balanced, symmetric example above, the five identical responses still yield only an 80% posterior.

Reality usually lies between independence and copying. Systems share some knowledge while independently performing parts of the reasoning. We should neither count every vote as independent nor infer that all incremental value is zero merely because training material overlaps. The relevant investigation concerns **where systems fail together and where they can correct one another** on the target tasks.

4.3 Majority accuracy differs from accuracy conditional on unanimity

With independent, symmetric systems of individual accuracy $p = 0.8$, the probability that a five-vote majority is correct is:

$$P(\text{correct majority}) = \sum_{k=3}^{5} \binom{5}{k} p^k (1-p)^{5-k} = 94.21\%.$$

A procedure that answers only when all five agree can achieve 99.90% accuracy, but unanimous answers occur on only 32.80% of questions. This is a change in coverage. Avoiding disputed questions can improve accuracy among answered questions without achieving that accuracy across the full task set. Any report of a unanimity benefit must also report how many questions were answered and how many were left unresolved.

5. A counterexample: more agreement, less reliability

5.1 Hold individual competence fixed and vary the common component

Construct a binary model as follows. On each question, with probability r, all five systems share a single judgment that is correct with probability p; they are therefore all right or all wrong. With

probability $1 - r$, the five answer independently, each with accuracy p. The mechanism is symmetric across the two truth states. The shared judgment is a mathematical device for representing common influence, not a description of an actual product architecture.

Each system's marginal accuracy remains p for every value of r. Pairwise correlation between their correctness indicators is exactly r. The probabilities of all being correct and all being incorrect are:

$$C_r = rp + (1 - r)p^5, \qquad W_r = r(1 - p) + (1 - r)(1 - p)^5.$$

Accuracy conditional on unanimity is therefore $C_r/(C_r + W_r)$, and unanimity coverage is $C_r + W_r$. With balanced priors, the posterior after unanimous endorsement of a claim also equals the former. Majority accuracy is $rp + (1 - r)P(\text{correct independent majority})$.

Table 1: Analytical illustration: each of five systems remains 80% accurate while the common component varies. Percentages are generated from one calculation file. They are not measurements of actual AIs.

Common r	Equivalent votes	Majority accuracy	Unanimity accuracy	Coverage
0.0	5.00	94.21%	99.90%	32.80%
0.1	3.57	92.79%	94.87%	39.52%
0.3	2.27	89.95%	88.63%	52.96%
0.6	1.47	85.68%	83.57%	73.12%
1.0	1.00	80.00%	80.00%	100.00%

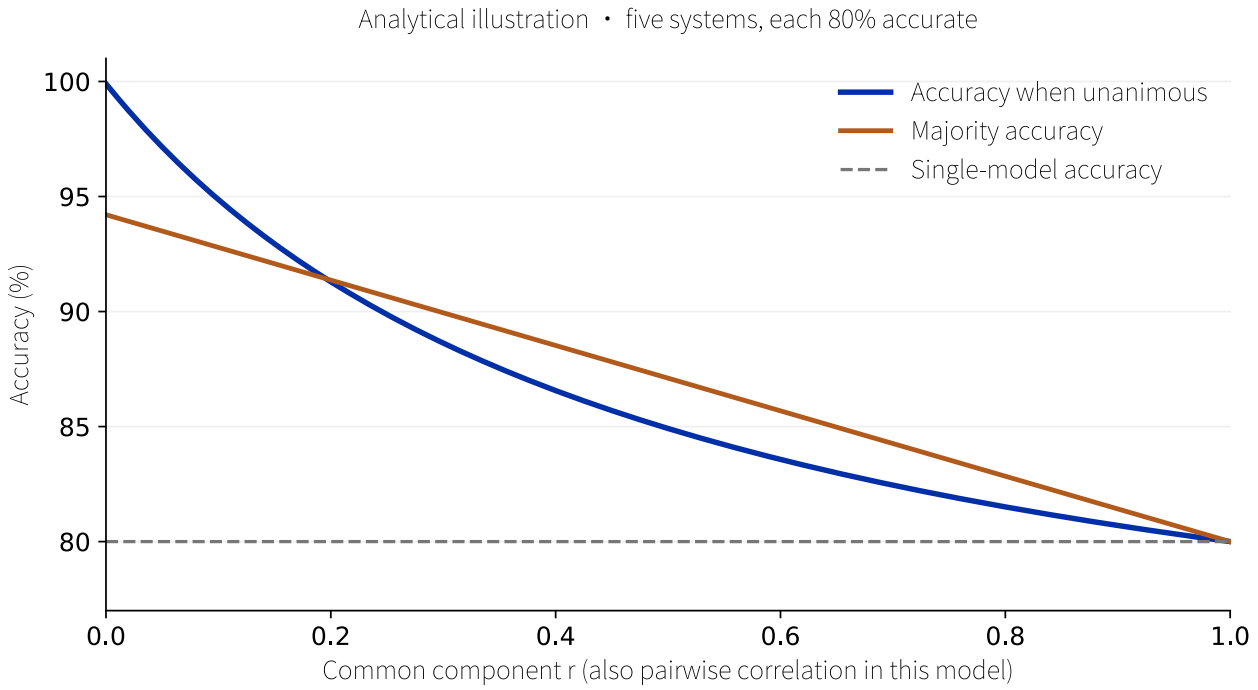


Figure 1: Analytical illustration: accuracy under unanimity and majority accuracy both fall as the common component increases. The curves have different conditioning denominators; unanimity coverage is reported separately in Table 1. Source: the paper's exact probability construction.

The striking feature is the movement in opposite directions. As r increases from 0 to 0.3, unanimity coverage rises from 32.80% to 52.96%, while accuracy conditional on unanimity falls from 99.90% to 88.63%. Consensus becomes easier to obtain and less reliable. A user observing only more frequent agreement could mistake increasing dependence for better verification.

Within this construction, for any fixed $r > 0$, increasing the number of systems indefinitely makes accuracy conditional on unanimity approach p. Both all-correct and all-incorrect events in the independent component become rare, so the remaining unanimous cases increasingly originate in the shared component. This limit depends on the construction and is not a universal law for correlated systems. It nevertheless identifies a testable mechanism: more systems can make selected unanimous cases more dominated by a common cause.

5.2 Effective votes are useful but cannot replace posterior probabilities

Let X_i indicate whether system i is correct. If all systems have accuracy p and equal pairwise correlation ρ, then:

$$\mathrm{Var}(\bar{X}) = \frac{p(1-p)}{n}[1 + (n-1)\rho], \qquad n_{\mathrm{eff}} = \frac{n}{1 + (n-1)\rho}.$$

The second expression defines an equivalent independent sample size **in terms of the variance of the average correctness indicator**. Five systems with $\rho = 0.3$ yield about 2.27. This is a useful warning about redundant information. It does not mean that 2.27 literal votes exist, and it does not justify replacing the exponent 5 in the independent unanimity formula with 2.27.

The reason is that means and pairwise correlations constrain only the first two moments, whereas unanimity is an event in the tail of the joint distribution. The appendix constructs two valid distributions with identical individual accuracy of 80% and pairwise correlation of 0.3, but accuracy conditional on unanimity of 88.63% and 94.29%. Estimating correlation therefore does not by itself establish the reliability of a particular consensus.

6. Four ways consensus becomes detached from evidence

6.1 One source is presented as several discoveries

Five AIs may cite five web pages that all repeat one report. They may cite different papers that reuse the same data or disputed measurement. Evidential dependence occurs at the levels of sources, data, and interpretation; it cannot be counted simply by URLs, institutions, or product brands.

A shared source is not inherently a problem. Several systems can reasonably agree by accurately interpreting a well-examined original dataset. The question is what each additional system checks: does it recalculate a result, verify a source, or seek counterexamples, or does it merely restate the same account? Different analyses of the same data can be valuable, but they must not be described as independent data collection.

Overlapping training material, similar optimization objectives, shared retrieval results, and similar task conventions are possible mechanisms of common error. These are candidate explanations, not assertions about any particular model's internals. Correlated errors may also reflect varying question difficulty or shared ambiguity. Correlation alone cannot identify which common cause is responsible.

6.2 Sycophancy changes what an endorsement means

Suppose an assistant becomes more willing to agree after hearing that the user has thought carefully and believes they are right. Agreement can then carry two kinds of information: a judgment about the proposition and adaptation to the response the user wants. Treating the second as the first exaggerates the strength of support.

An extreme analytical example makes the distinction clear. Suppose a system endorses both true and false user claims with probability 90%. Five independent systems may all nod, but the likelihood ratio is 1, so their endorsements do not improve truth assessment. This is not an

estimated real-world sycophancy rate. It illustrates why frequent endorsement is not equivalent to discrimination between true and false claims.

Respect for a user does not require sycophancy. A good assistant can acknowledge emotions and motivations while identifying an error. It can also reasonably revise a conclusion when the user supplies new facts. Testing sycophancy requires holding factual material constant while varying expressed positions, rather than treating every revision as a failure.

6.3 Conversation transmits reasons and can also spread errors

If five systems answer independently and then discuss, a valid proof found by one may be adopted by the others. Such influence is beneficial. However, the final agreement follows information exchange and must not be counted as five independent discoveries.

A clear but mistaken argument can instead become a common anchor. In human estimation tasks, Lorenz and colleagues found that social influence could narrow differences without improving accuracy and could raise confidence. This is not direct experimental evidence about AI. It does explain why convergence and correctness should be measured separately. [8]

Discussion should be judged by errors corrected, valid objections preserved, and verifiable material added, not by whether everyone eventually adopts one position. Huang and colleagues' work on intrinsic self-correction similarly motivates distinguishing reconsideration without external feedback from revision informed by new verification signals. Its findings about particular models must not become a permanent denial of future self-correction capabilities. [9]

6.4 Showing only the final five endorsements creates selected consensus

If a user changes prompts, switches models, or regenerates responses until five endorsements appear, then deletes earlier objections and abstentions, the reported event has changed. It is no longer that five systems fixed in advance all endorsed the claim. It is that at least one of several attempts produced unanimous endorsement.

In the $r = 0.3$ example in Table 1, a false proposition receives unanimous endorsement from one five-system panel with probability 6.0224%. Assuming panels are independent across attempts, 20 attempts produce at least one false consensus with probability $1 - (1 - 0.060224)^{20} = 71.13\%$. This is a stipulated calculation illustrating selection. Real repeated queries may be dependent, so the number must not be transferred directly to them.

The point is not that every optional stopping rule invalidates Bayesian inference. Coherent updating is possible when the full sequence is retained and sampling and reporting mechanisms are correctly modeled. The problem is **hiding unsuccessful attempts while interpreting the selected result as one fixed draw**. Keeping objections and query histories therefore has epistemic significance, beyond administrative tidiness.

7. From machine opinions to human judgment

7.1 Five fluent arguments can persuade without adding verification

When a conclusion is expressed in technical, economic, historical, and ethical language, it may appear to have passed tests across several fields. Diversity of expression is not necessarily diversity of evidence. We must ask whether the arguments support the same precise claim, whether each establishes only a weaker version, and whether they were constructed backward from the desired conclusion.

Turpin and colleagues found that reasoning explanations from the models they tested did not always faithfully report biases influencing the answer. An explanation can therefore be plausible

without being a complete account of how the output was produced. This motivates checking verifiable steps instead of treating richer wording as proof of independent mechanisms. [10]

There can also be a transfer of responsibility: “It is not merely my judgment; five AIs approve.” Yet a person or system designer still decides how to formulate the question, which models to select, which answers to retain, and when to stop asking. Consensus does not erase those choices or replace responsibility for the resulting decision.

7.2 Seeking truth may be less comfortable than receiving agreement

Asking users to record an initial judgment, examine primary evidence, and explain acceptance or rejection of an AI answer can be more demanding than displaying a conclusion. Buçinca and colleagues found that cognitive forcing interventions reduced overreliance in their experiment, but the most effective designs were not necessarily the most positively rated. This supports evaluating user experience separately from judgment quality. [11]

Greater confidence alone therefore cannot establish that a tool works well. A system that makes users accept incorrect answers faster may score highly on fluency and satisfaction while causing epistemic harm. Conversely, identifying an unresolved central issue can reduce confidence immediately while making later judgment more reliable.

7.3 Minority opinions deserve examination, not romanticization

Disagreeing with a majority does not automatically bring one closer to truth. An incorrect objection gains no truth merely by being bold. What deserves preservation is a specific, evidence-supported objection that has not been answered. Assigning a system to play a contrarian can also produce arguments that oppose for the sake of opposing.

Hong and Page’s formal model shows the value of functional problem-solving diversity under particular conditions. Applied to AI, the relevant search is for different verification paths, knowledge coverage, and failure patterns, not merely different persona labels. Competence and complementarity need to be considered together. [12]

If a minority system produces an executable counterexample while the majority offers assertions, the counterexample deserves greater weight. If the majority supplies a reproducible proof and the minority merely disagrees, there is no reason to split the difference mechanically. Respecting disagreement means allowing evidence to change the conclusion.

8. A practical verification procedure

The analysis motivates the following procedure. These recommendations have theoretical and literature support, but the combined procedure has not been evaluated as a complete intervention.

First, fix the claim and its failure conditions. Replace “this is a good plan” with an assessable proposition, such as “under the specified inputs and constraints, this algorithm always produces a valid result.” State in advance which record, counterexample, or calculation would defeat it, so that the object of agreement does not quietly shift during discussion.

Second, preserve initial answers uninfluenced by other answers. Begin with prompts that are as neutral as practicable and supply the necessary material. Ask systems separately for conclusions, evidence, conditions, and uncertainty. Do not initially display the user’s desired conclusion or other systems’ endorsements. When that information is necessary, record what difference it makes.

Third, trace evidence to common origins. Record the original source, data version, and check performed for each major reason. If five URLs depend on one original source, connect them to the

same evidence node. Data provenance, calculation, and interpretation can be checked separately without counting one discovery several times.

Fourth, use different roles to produce different checks. One system can verify sources, another recalculate, another search for counterexamples, another examine scope, and another organize unresolved conflicts. This does not guarantee independence, but it has a better chance of producing complementary information than five speaking styles. If only two checks are useful, there is no need to invoke five systems simply to fill a quota.

Fifth, allow unresolved outcomes. Each revision should identify the new evidence, calculation, or corrected premise that motivated it. Unresolved disputes belong in the final record. Fluency, assigned job titles, and self-reported confidence cannot independently serve as the final arbiter.

Sixth, calibrate on comparable questions with known answers before deciding how to act. Record the combination's accuracy, error rate under unanimity, abstention coverage, and important failure types. Test whether high confidence corresponds to low error. When calibration is insufficient, "several supporting responses obtained; independence and error rates unknown" is more honest than an exact posterior claim.

A fictional example illustrates the procedure. Five assistants declare that an algorithm works for every input, but all cite one tutorial. Another check constructs a boundary input, and running the program produces a result that violates the stated constraint. The reproducible counterexample and clarification of the tutorial's scope advance understanding. How many systems subsequently accept the counterexample is a later propagation outcome. If the environment or the test itself is flawed, the counterexample also needs verification.

Verification has costs. A low-cost, reversible choice may not warrant all six steps; a consequential judgment that is difficult to reverse demands stronger external checks. Action depends on the loss from error, the cost of delay, and reversibility, not confidence alone. Whether another query is worthwhile can be framed as whether its expected reduction in decision loss exceeds the cost of checking. This is a decision framework, not a universal threshold measured by this paper.

9. How could the argument be challenged?

The following experiments have not been conducted for this paper. Future tests should fix tasks, model versions, prompts, call and time budgets in advance, preserving every output and abstention. Factual tasks, formal reasoning, and open-ended evaluation should be analyzed separately rather than collapsed into one average score.

At least five procedures should be compared: a strong single model checking within the same budget; repeated sampling from one model; independent voting across models; independent answers followed by discussion; and the evidence-checking division of work in Section 8. Total cost and available tools must be controlled. Otherwise, an apparent multi-system benefit may simply reflect more computation or additional retrieval. Tasks and model selection should be fixed before evaluation rather than reporting only favorable combinations afterward.

The same factual material can be paired with neutral, supportive, and opposing user positions to test sensitivity to attitude. Preserve independent initial and post-discussion answers to distinguish propagation of correct views, propagation of errors, and convergence without improvement. Use separate calibration and held-out test sets when weights or combinations require tuning.

Primary outcomes should include final accuracy; error rate under unanimity and corresponding coverage; adoption of correct minority views; acceptance of erroneous citations; and correction benefit per unit cost. Numerical confidence outputs additionally permit calibration assessment and

Brier scores. A separate future study of human judgment would require appropriate participant authorization before measuring accuracy and confidence before and after advice. No participants were recruited for this draft.

Statistical units must match the question. Repeated calls and different prompts are often nested within the same item and must not all be treated as independent observations. Paired comparisons and resampling by item can be used, while uncertainty from model selection should be stated. For disputed human labels, retain annotation distributions rather than unconditionally naming the majority label factual truth.

These comparisons allow real failure. If independent voting consistently beats the more elaborate verification procedure under fair budgets, expectations about procedural complexity should be reduced. If discussion without new external material reliably improves results, research should examine how it makes better use of knowledge already present; “no new source, no progress” would be an unjustified position. If expressed user attitudes have no effect on a calibrated system, general sycophancy research should not be used to penalize it. The framework should locate benefits and failures, not protect a predetermined verdict.

10. Limitations

First, this is a theoretical analysis and targeted synthesis, not a measurement of the five AIs a reader happens to use. Individual accuracy, common-component strength, and priors are stipulated to explain mechanisms. No product’s reliability can be read from the charts.

Second, the binary, symmetric, equal-competence model simplifies reality. Multiple-choice tasks have several possible wrong answers. Free-form outputs introduce semantic equivalence, partial correctness, and omissions. Measuring agreement can itself be erroneous. These complexities are not estimated here.

Third, the literature selection is not exhaustive. Some sources were checked only at the original-abstract level, without auditing data and every appendix. Studies use different models, benchmarks, tools, and dates, so their results cannot be spliced into a single effect estimate. Recent preprints especially require further scrutiny.

Fourth, independence is not the only virtue. Shared high-quality evidence, reasonable discussion, and convergence on a valid proof can all help. The concern is repeated counting of agreement generated through sharing, not a demand that systems remain isolated or that more objections are always better.

Fifth, traceable evidence is not infallible. Original papers, test programs, and human judgments can fail. This draft was itself produced with AI assistance. Executed calculations constrain numerical error but cannot replace independent research judgment or external peer review.

11. Conclusion: turn agreement back into inspectable reasons

When five AIs agree with me, I may indeed be closer to the truth. That requires endorsements that discriminate between true and false claims, genuine evidential increments not already explained by existing information, appropriate treatment of correlated errors and selection, and conclusions open to external checks.

I may also merely feel more certain that I am right. Five writing styles, five brands, or five rounds of discussion cannot by themselves exclude this possibility. Consensus without additional checking may expand the circulation of a reason without expanding what it establishes.

The title ultimately asks us to shift attention from how many intelligent systems are on our side to which reasons should make us revise our position. Sometimes the answer is five complementary checks; sometimes it is one reproducible counterexample; sometimes it is an acknowledgment that the available information is insufficient.

Progress toward truth is marked by judgments that better withstand criticism and change when the evidence changes. Only within such a procedure can we determine what agreement from five AIs is worth.

Appendix A. Equal correlation, different reliability under unanimity

Let K be the number of correct systems among five. Sample K from a distribution, then choose uniformly which K systems are correct. This produces an exchangeable joint distribution. In Table 2, A is the Section 5 construction with $p = 0.8, r = 0.3$. B adds $0.03 \times (-1, 5/3, 0, 0, -5/3, 1)$ to the six probabilities in A.

Table 2: Analytical counterexample. Both columns are valid probability distributions; rows indicate the number of correct systems, not observed sample frequencies.

K	Distribution A	Distribution B
0	0.060224	0.030224
1	0.004480	0.054480
2	0.035840	0.035840
3	0.143360	0.143360
4	0.286720	0.236720
5	0.469376	0.499376

Each column sums to 1, with $E[K] = 4$ and $E[K(K-1)] = 13.76$. Individual accuracy is therefore $4/5 = 0.8$, the pairwise probability of being correct is $13.76/20 = 0.688$, and pairwise correlation is $(0.688 - 0.8^2)/(0.8 \times 0.2) = 0.3$. Yet $P(K = 5 \mid K \in \{0, 5\})$ is 88.63% for A and 94.29% for B. The reproduction script checks nonnegative probabilities, normalization, and both moments exactly.

Materials, licensing, and reproduction

The probability constructions require no model API, participant data, or random sampling. Sections 4–5 specify the model and formulas, and Appendix A lists every counterexample probability. Exact-fraction enumeration of all 32 patterns of five binary systems verifies normalization, individual means, pairwise correlations, and accuracy conditional on unanimity. These calculations were actually executed. Calculation code, analytical data, original figures, the shared bibliography, evidence register, glossary, and revision records are maintained in the local working project. The default public release consists of the complete Chinese and English PDFs and the paper introduction.

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AI use statement

On October 26, 2026, AI assistance in the Codex environment supported searching, argument organization, drafting, English translation, coding, and layout. The session identifies its model as GPT-6, but the exact server snapshot cannot be independently verified, so no more specific model designation is reported. Major sources were checked against original web pages; values were actually recomputed using exact fractions and enumeration; the two texts were checked for correspondence. The scope of HTML and PDF inspection is recorded in the local verification records.

Revisions and self-checks by the same assistant are not counted as independent review. The model experiments reported by cited authors were not rerun for this manuscript. All unperformed studies are described only as future research designs.

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